Confound Auditing in Medical Classifiers: Evidence from TB, Glioma, and Malaria

Rez Samantha Z. Floresca, Ateneo de Manila, rez.samantha.floresca@student.ateneo.edu
Edric Castel C. Hao, Analog Devices, edriccastel.hao@analog.com

Abstract

A single aggregate performance number provides limited evidence of whether a medical classifier has learned disease or a correlate of it. We propose an iterative falsification audit for frozen-feature classifiers in which each verdict determines the next test. The audit combines a confound-stratified test that ablates shortcut and residual directions under random-direction and retrained-head controls, decision rules that assign a follow-up test to each outcome, and a planted-marker calibration that estimates the false-flag rate and power. On the planted marker, 0 of 20 splits were flagged when no association was present. The planted marker was detected in at least 85% of splits for two of three encoders when its prevalence among tuberculosis (TB) cases was at least 0.7, with lower sensitivity for the TorchXRayVision (XRV) encoder. We applied the audit to three disease settings built from public data. In TB radiographs, extrapulmonary involvement within TB cases was not supported as a shortcut, though XRV normal-class accuracy fell. Acquisition source was highly decodable (AUC 0.999 to 1.000) and perfectly aligned with the disease label in the training data. A classifier trained on Belarus TB cases and Radiological Society of North America (RSNA) normals achieved AUCs of 0.455 to 0.542, near or below chance, on source-matched Shenzhen and Montgomery data, and nonlinear probes still decoded source after linear erasure (AUC 0.941 to 0.998). In glioma, ablating World Health Organization (WHO) grade collapsed predictions to a single class, motivating grade-excluded refits that achieved discordant AUCs of 0.563 to 0.812 across imputation choices. In malaria, replacing estimated parasite pixels did not eliminate predictive performance (patient-grouped AUC 0.952 to 0.977), and uninfected cells retained information about patient infection status (AUC 0.946 to 0.954).

1. Introduction

Medical image classifiers can reach high AUC through features that correlate with the label but carry no diagnostic meaning (Zech et al., 2018; DeGrave et al., 2021; Geirhos et al., 2020). Ruling out one shortcut says nothing about the others, and confirming one does not tell us whether disease signal is left. We use *confound* for a label-aligned variable and *shortcut* for one the classifier relies on instead of disease signal.

Gap and purpose. Most prior work tests one named shortcut or locates a cue once suspected (Section 2). We know of no protocol that states what each outcome of a shortcut test should lead to next, reports the audit's own false-flag rate and power, or checks for nonlinear leakage after linear erasure. This study proposes an *iterative falsification audit* in which each verdict selects the next test (Section 3), and examines how it behaves on three disease settings built from public data where the shortcut is not known in advance.

Contributions. We propose and evaluate, in medical images, a shortcut audit that reports its own false-flag rate and power on a planted marker of controlled strength, across three frozen TB encoders (Section 5.1). We show that linear erasure of a label-coincident source confound leaves it decodable by nonlinear probes (Section 5.2), and that erasure directions learned in one encoder do not transfer to another (Appendix J). We apply the audit to three case studies: a Belarus/RSNA TB classifier on source-matched data (Section 5.2), a glioma grade ablation that reduces to a constant prediction (Section 5.3), and patient infection status predicted from uninfected NIH malaria cells (Section 5.4). Our contribution is a calibrated, iterative audit that quantifies its own performance (Appendix C).

2. Related Work

Reported shortcuts in medical imaging include site identity (Zech et al., 2018), patient attributes (Jabbour et al., 2020), chest drains (Jiménez-Sánchez et al., 2023), and regions outside the lungs (Maguolo and Nanni, 2021). Additionally, error auditing can find hidden low-performing subsets without a prior hypothesis (Oakden-Rayner et al., 2020). TB classifiers have also been shown to lose accuracy on other populations (Sathitratanacheewin et al.,

2020). Other studies link attribute encoding to unfairness (Brown et al., 2023), use layer-wise probes to locate drain and scanner cues (Pedersen et al., 2026), and test reliance on off-target findings through synthetic edits (Cooke et al., 2025). In brain MRI, iterative nullspace projection has been used to reduce site-related signals in embeddings (Rahbar, 2026).

These approaches leave open the question of whether erasing a linearly encoded cue eliminates the information itself. Iterative nullspace projection (INLP; Ravfogel et al., 2020) and Least-squares Concept Erasure (LEACE; Belrose et al., 2023) target concepts detectable by linear probes, but ablating a probe direction does not establish that the corresponding information is necessary for prediction, since a refitted probe may recover it (Jiang, 2026). Last-layer retraining can recover core features even when standard training encourages reliance on shortcuts (Kirichenko et al., 2023).

Most of these studies test a named or suspected cue. We found none that calibrates an audit on a planted shortcut, checks erasure with nonlinear probes or across encoders, evaluates a TB classifier on source-matched data, or tests patient state in uninfected malaria cells (gaps G1 to G8, Appendix C).

3. Audit Procedure

Procedure. The audit names a candidate shortcut s, stratifies held-out samples by concordance with it, ablates shortcut and residual directions under three controls, and uses the verdict to select a targeted follow-up. A planted marker calibrates the audit itself (Figure 2, Appendix B).

Stratified audit. Let z be a frozen feature vector (an encoder output, or the input variables for tabular glioma data), y the label, and s a binary candidate shortcut. Held-out samples are *concordant* when s agrees with y as it does in training and *discordant* otherwise, and a classifier that relies on s scores lower on discordant samples. A lower discordant score is consistent with reliance but can also reflect subgroup difficulty. We fit a logistic classifier with weight vector d_c and a logistic probe for s with weight vector d_s. The *residual direction* d_r is the part of d_c orthogonal to d_s. *Ablating* a unit direction $\hat{d}$ replaces z by $z - (z^\top \hat{d})\hat{d}$.

Controls. Because d_r is built from the classifier, ablating it is expected to damage that classifier. Every ablation is therefore reported with the original head, with a head retrained on ablated features, and against 200 covariance-matched random directions, which gives an empirical p-value. For a confound with known labels (TB source) we apply LEACE and up to 200 rounds of INLP, then probe held-out data with linear, multilayer perceptron (MLP), and histogram gradient-boosting (HGB) probes.

Decision rules. If the tested shortcut is not supported, we search for other variables aligned with the label (source, patient). If it is supported, we test whether disease signal survives without it (a *refit* with the shortcut variable excluded from the inputs). If a restricted view stays predictive, we add views and groupings that separate competing explanations (random crop, grayscale, patient-grouped splits). A negative verdict rules out only the tested form of reliance under the conditions the calibration covers.

Calibration and reporting. We plant a bright corner block with a soft gradient in Shenzhen and Montgomery radiographs with $P(\text{marker} \mid \text{TB}) = \rho$, where ρ is the marker prevalence among TB cases and $1 - \rho$ the prevalence among normals, so $\rho = 0.5$ is the no-association setting and $\rho = 1$ gives perfect association. Over 20 splits per ρ the audit flags reliance when LEACE on the recovered marker direction raises AUC on marker-flipped test images by more than 0.02 and above the 95th percentile of random directions. Bracketed ranges are 95% bootstrap intervals over test samples (patients in the patient-state tests, Appendix H) and exclude training variance.

4. Data and Setup

TB. DICOM cases from Belarus TB Portals (Rosenthal et al., 2017), 517 pulmonary-only (group A) and 533 with extrapulmonary disease (group B), are paired with 8,851 normals from the Radiological Society of North America (RSNA) Pneumonia Challenge (Shih et al., 2019). We call this training population Belarus/RSNA. The candidate shortcut is extrapulmonary involvement (group A vs. B). Shenzhen and Montgomery radiographs (Jaeger et al., 2014), where TB and normals share an acquisition pipeline, provide a source-matched evaluation. TB features come from three frozen encoders: the DenseNet-121 (Huang et al., 2017) from TorchXRayVision (Cohen et al., 2022) (XRV, 1024-d; its training pool includes RSNA images,

Appendix L), ResNet-50 (He et al., 2016), and DINOv2-small (Oquab et al., 2024), which we call DINOv2. XRV is also run on harmonized images.

Glioma and malaria. Tabular TCGA data (Ceccarelli et al., 2016) from cBioPortal (Cerami et al., 2012; Gao et al., 2013) cover 995 patients (1,122 rows minus 127 without isocitrate dehydrogenase (IDH) status). The candidate shortcut is World Health Organization (WHO) grade as recorded in TCGA, and MGMT (O6-methylguanine-DNA methyltransferase), 1p/19q, and TERT (telomerase reverse transcriptase) are the other inputs. TCGA labels predate the 2021 WHO classification (Louis et al., 2021, Appendix D). The test set has 257 concordant and 141 discordant (IDH-wildtype lower-grade or IDH-mutant glioblastoma) patients. Malaria images are National Institutes of Health (NIH) thin-smear cells (Rajaraman et al., 2018), encoded by ResNet-50, DenseNet-121, and DINOv2 on six views: the full image, the parasite region, the red blood cell (RBC) view, a random off-center crop, grayscale, and the cell silhouette. The RBC view replaces estimated parasite pixels with the median cell color. Evaluation is 5-fold cross-validation grouped by 201 patient identifiers parsed from file names, one more than the 200 patients the dataset describes (patient-grouped splits, Appendix D).

5. Results

5.1. Calibration on a planted marker

At $\rho = 0.5$ the audit never flagged reliance (0 of 20 splits for each encoder, so the one-sided 95% upper bound on the false-flag rate is 0.14). For DINOv2 and ResNet-50 the flag rate was 0.85 at $\rho = 0.7$ and 1.00 at $\rho = 0.8$ and 0.9 (Figure 1c). The main TB encoder, XRV, had lower power, with flag rates of 0.30, 0.75 and 0.70 at $\rho = 0.7$, 0.8 and 0.9, and 0.00 at $\rho = 1.0$. The recovered direction had cosine 0.64 to 0.72 with the planted one for DINOv2 and ResNet-50 and about 0.25 for XRV (Appendix I), so a negative audit on XRV features does not exclude reliance, and XRV power is not monotone in ρ.

5.2. TB: extrapulmonary involvement and acquisition source

Extrapulmonary. The extrapulmonary direction was fit within TB-positive cases, so the test cannot show whether extrapulmonary involvement separates TB from normal. In XRV and DINOv2 features, ablating it left positive-class accuracy unchanged, so extent was not supported as a shortcut within cases. In XRV it lowered normal-class accuracy from 0.998 to 0.835 ($p = 0.01$ against random directions), and a retrained head restored 0.998. The ablation does not center the features, so it shifts logits by a constant. A threshold shift can therefore produce this pattern, and head recovery does not show that the direction is dispensable. Residual-direction ablation collapsed positive-class accuracy (to 0.108 and 0.236 in XRV), and a retrained head recovered it. Both groups are Belarus DICOM, so this test cannot detect reliance on format.

Source. Source was decodable at AUC 1.000 (DINOv2, ResNet-50) and 0.999 (XRV). Every Belarus image is TB-positive and every RSNA image normal, so source and disease label are perfectly confounded in training. The Belarus/RSNA-trained classifier scored AUC 0.455 [0.417, 0.499], 0.511 [0.471, 0.552], and 0.542 [0.503, 0.584] for DINOv2, ResNet-50, and XRV on combined Shenzhen and Montgomery data (Figure 1b). Because these datasets also differ in population from Belarus and RSNA, the performance drop cannot be attributed to format alone (Appendix F). Harmonizing format raised the classifier's AUC to 0.752 [0.715, 0.785] on source-matched data, while source remained decodable (AUC 0.999). Image metadata alone reached AUC 0.748 within the source-matched data, so matching left label-aligned variables.

Erasure. LEACE reduced linear source-probe AUCs to 0.470 to 0.494 across feature sets, near chance, in one step. MLP probes still achieved AUCs of 0.989 to 0.998, while HGB probes achieved 0.985 to 1.000. INLP required 100 to 200 rounds to bring linear probes to chance, yet MLP AUCs after 200 rounds remained at 0.941 to 0.994 (Figure 1a, Appendix F). A linear-only audit would conclude that source was erased, even though nonlinear probes still decode it. Source-matched TB classification was unchanged by erasure (XRV AUC 0.808 to 0.825).

5.3. Glioma: grade ablation reduces to a constant

With grade as input, the classifier reached concordant AUC 1.000 and discordant AUC 0.000, as implied by the subgroup definition, and the grade direction had cosine 1.00 with the classifier direction, leaving no

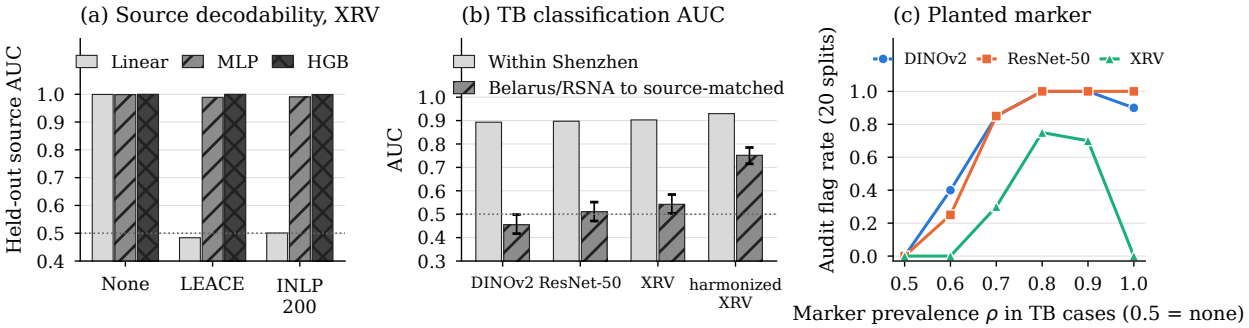


Figure 1: (a) Held-out source-probe AUC after erasure of source in XRV features (chance 0.5). (b) TB classification AUC within Shenzhen (cross-validated) and for the Belarus/RSNA-trained classifier on source-matched data, with 95% bootstrap intervals. (c) Fraction of 20 splits in which the audit flags reliance on a planted marker, by marker prevalence ρ in TB cases.

residual direction. Ablating it made the classifier predict IDH-mutant for every patient (accuracy 0.490 on concordant patients, 0.000 on IDH-wildtype lower-grade patients, and 1.000 on IDH-mutant glioblastoma patients). Subgroup accuracy therefore reflects the class prior. The ablation shows that the classifier depends on the grade direction, but not that this dependence is spurious, because grade and IDH status are related in the WHO classification.

The glioma conclusion rests on the grade-excluded refit. With zero imputation of missing values, it reached concordant AUC 0.877 (SD 0.014 over 20 splits) and discordant AUC 0.738 [0.649, 0.823]. Median imputation gave 0.812, median imputation with missingness flags gave 0.563, dropping 1p/19q gave 0.624, and missingness indicators alone gave an overall AUC of 0.666. Co-deletion of 1p/19q implies IDH mutation by definition (Louis et al., 2021), so part of the grade-independent signal is definitional. We conclude only that an IDH signal remains without grade and that its size is unstable (discordant AUC 0.563 to 0.812, Appendix G).

5.4. Malaria: signal after parasite-pixel replacement, including patient state

Patient-grouped AUC was close to cell-level AUC (0.977 vs. 0.982 for ResNet-50), so patient overlap does not explain the high values (Appendix H).

Staining. A saturation-defined staining ablation left ResNet-50 accuracy at 0.927 against 0.928 ($p = 0.95$), and grayscale (0.975 to 0.989) and color-only features (0.967 to 0.969) stayed near the full image (0.977 to 0.991). Neither isolates a stain effect.

RBC view. The RBC view reached AUC 0.952 to 0.977 against 0.811 to 0.848 for the random crop. Neither comparison locates the signal: the RBC view's flat patch can itself mark infected cells, and the crop differs in size and position.

Patient state. Patient infection status was predictable from uninfected cells alone at patient-level AUC (scores averaged per patient) 0.946 to 0.954, and a probe identified the correct patient among 201 at top-1 accuracy 0.50 to 0.58 (chance 0.005, Appendix H). These results fit a slide-level or patient-level cue, which cell-level splits can leak (Hou et al., 2026), but the lack of slide metadata prevents identifying it. Unannotated parasites or infection-related morphology in uninfected cells could produce the same pattern.

6. Conclusion

The iterative falsification audit assigns a follow-up test to each outcome. On a planted marker the audit flagged 0 of 20 splits at no association and at least 85% of splits at prevalence 0.7 or higher for DINOv2 and ResNet-50. Across three disease settings built from public data, follow-up tests showed that a Belarus/RSNA TB classifier performed near or below chance on source-matched data and that source remained decodable after linear erasure. The grade-excluded glioma signal depended on imputation, and uninfected malaria cells appeared to carry patient infection status. The calibration used one synthetic marker and one flag criterion (LEACE gain), so other criteria, shortcut types, encoders, and domains remain to be calibrated (Appendix L).

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Appendix A. Statements

Data and Code Availability. All data are public and de-identified: Shenzhen and Montgomery chest radiographs (Jaeger et al., 2014), the Belarus TB Portals collection (Rosenthal et al., 2017) (Kaggle mirror), RSNA Pneumonia Challenge normals (Shih et al., 2019), the TCGA LGG/GBM cohort (Ceccarelli et al., 2016) via cBioPortal (Cerami et al., 2012; Gao et al., 2013), and the NIH NLM malaria cell images (Rajaraman et al., 2018). No new data were collected.

IRB. Secondary analysis of public de-identified datasets, not human subjects research requiring IRB approval.

Appendix B. Audit Procedure Figure

Figure 2 summarizes the audit described in Section 3.

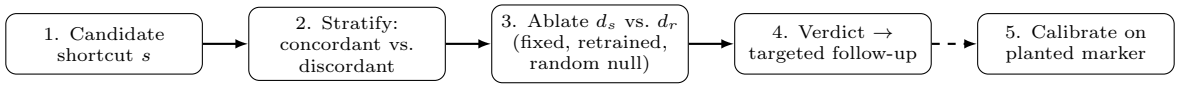


Figure 2: The audit. Step 3 reports every ablation with a retrained head and a random-direction null, step 4 selects the follow-up for each domain (Section 5), and step 5 calibrates the audit itself.

Appendix C. Literature Gaps, Contributions and Novelty

Gaps in the literature. The gaps below are stated against the closest work we found, and each means that we found no study that does what is described.

G1. *No calibration of medical-imaging shortcut audits.* Shortcut tests, probes, and counterfactual edits have been used for chest radiographs (Jiménez-Sánchez et al., 2023; Pedersen et al., 2026; Cooke et al., 2025; Brown et al., 2023), and semi-synthetic testbeds exist for explanation methods (Adebayo et al., 2022), but none measures the false-flag rate and detection power of a representation-level audit across shortcut strengths and encoders in medical imaging.

G2. *No check of linear erasure with nonlinear probes for a label-coincident medical-imaging source confound.* Site or source signal has been removed by null-space projection in brain-MRI embeddings (Rahbar, 2026), and ablating a probe direction is known not to test necessity in language-model interpretability (Jiang, 2026). No radiograph study applies LEACE and INLP to a source confound that coincides with the label and then probes the residual with nonlinear models.

G3. *No eraser transfer across encoders in medical imaging.* No study fits an eraser in one frozen medical-imaging encoder and tests it in another.

G4. *No source-matched evaluation of a Belarus/RSNA-trained detector.* Dataset specificity of TB detectors is known (Sathitratanacheewin et al., 2020), and source identification has been studied for other chest radiograph sets (Dack and Dai, 2025). No evaluation tests a detector built from Belarus TB Portals cases (Rosenthal et al., 2017) and RSNA normals (Shih et al., 2019) on data where both classes share a source.

G5. *No patient-state test beyond slide-disjoint splits in NIH malaria.* Slide-disjoint evaluation lowers reported accuracy (Hou et al., 2026). No test asks whether uninfected cells carry patient infection status, or evaluates parasite-removed views under patient-grouped splits.

G6. *No shortcut audit of grade in tabular IDH prediction.* The WHO classification defines IDH-based entities (Louis et al., 2021). No study audits grade as a shortcut for IDH prediction on TCGA and reports that ablating it collapses to a constant.

G7. *No CKA check after erasing a known medical confound.* Centered kernel alignment (CKA) (Kornblith et al., 2019) is interpreted through linear decoders (Harvey et al., 2024) and has known reliability problems (Davari et al., 2023). No study reports it before and after erasing a known imaging confound.

G8. *No functional-use control combined with a planted shortcut.* Restorability of shortcuts after unlearning is tested on natural-image and dermatology benchmarks (Lu and Ji, 2026), and calibration shifts under shortcuts are studied for chest drains (Kina and Petersen, 2026). No audit combines fixed head, retrained head and random-direction nulls for every ablation and validates it against a planted shortcut.

Contributions.

C1. We report the false-flag rate (0 of 20 splits at $\rho = 0.5$ for three encoders) and detection power (flag rate 0.85 to 1.00 at $\rho \geq 0.7$ for DINOv2 and ResNet-50, and 0.30 to 0.75 for XRV at $\rho = 0.7$ to 0.9) of the representation-level audit against a planted marker in medical images, and show that the audit can miss a perfectly associated planted marker (XRV at $\rho = 1.0$).

C2. We evaluate a Belarus/RSNA-trained TB classifier on source-matched Shenzhen and Montgomery data, where it scored AUC 0.455 to 0.542 although within-source classification was 0.844 to 0.903, and show that format harmonization lifted it only partly, to 0.752.

C3. We show, for a medical-imaging source confound that coincides with the label, that linear erasure (LEACE, 200-round INLP) leaves source decodable by MLP (0.941 to 0.998) and HGB probes, with source-matched classification unchanged.

C4. We test eraser transport and head inheritance across frozen medical-imaging encoders, and report that source decodability by an MLP stays at 0.991 or above under every transported eraser (Appendix J).

C5. We report CKA before and after erasing a known imaging confound: CKA with the original encoder stays at 0.890 to 0.978 while linear source decodability falls to chance.

C6. We test, in NIH malaria cell images, whether uninfected cells from patients with an infected cell differ from uninfected cells of patients with none (AUC 0.612 to 0.675) and whether patient infection status can be predicted from uninfected cells (AUC 0.946 to 0.954).

C7. We report that ablating WHO grade in TCGA tabular IDH prediction collapses predictions to a constant, and that the grade-excluded refit ranges from 0.563 to 0.812 across imputation choices.

The audit also reports every ablation with fixed-head, retrained-head, and random-direction controls and records its deviations (Appendix K). These controls combine known tools.

What we do not claim. We do not claim novelty for shortcuts in TB, malaria, or glioma data, for the component methods (direction ablation, probing, LEACE, INLP, CKA, marker planting), or for slide leakage in NIH malaria (Hou et al., 2026). The audit does not establish the underlying causes of the observed performance patterns. It also leaves open whether format alone explains the TB source difference and whether malaria staining or localization drives the malaria results.

Appendix D. Procedure Details

Data. *TB.* Belarus TB Portals DICOM cases (517 pulmonary-only, 533 with extrapulmonary disease) and 8,851 RSNA normals form the training population. Shenzhen and Montgomery radiographs, in which TB and normals share an acquisition pipeline, form the source-matched evaluation. *Glioma.* Tabular TCGA data for 995 patients with IDH status, with WHO grade, MGMT (O6-methylguanine-DNA methyltransferase), 1p/19q and TERT (telomerase reverse transcriptase) as inputs (11.6%, 2.1% and 67% missing for the last three). The concordant test set has 257 patients and the discordant set 141 (104 IDH-wildtype lower-grade, 37 IDH-mutant glioblastoma). Grade and IDH labels follow TCGA, which predates the 2021 WHO classification. Under the 2021 classification IDH-mutant tumors are no longer called glioblastoma and some IDH-wildtype lower-grade tumors count as glioblastoma, so the discordant subgroups follow TCGA labels and not current diagnostic criteria. *Malaria.* NIH thin-smear cell images with 201 patient groups parsed from file names, median 85 cells per group, 151 groups with at least one infected cell, 10,364 uninfected cells from those patients, and 3,415 from the others. *Patient-grouped* splits keep all cells of a patient in one fold, *cell-level* splits do not, and *patient-level* AUC averages scores per patient.

Features. Frozen encoders, all run on 224×224 inputs: for TB, the TorchXRayVision DenseNet-121 (XRV, 1024-d) with weights `densenet121-res224-all` (Cohen et al., 2022), torchvision ResNet-50 (`IMAGENET1K_V1`), and DINOv2-small (`facebook/dinov2-small`); for malaria, torchvision ResNet-50 and DenseNet-121 (both `IMAGENET1K_V1`, so the malaria DenseNet-121 is not the XRV model) and DINOv2-small. Glioma uses the input variables directly. Harmonization applies histogram equalization followed by a JPEG quality-90 round trip to all TB sources before feature extraction.

Classifiers and probes. Heads and shortcut probes are logistic regressions. Source probes after erasure use logistic regression, an MLP with one hidden layer of 256 units and early stopping, and a histogram gradient-boosting (HGB) classifier with 150 iterations, all implemented in scikit-learn (Pedregosa et al., 2011). Erasers are fit on 60% of the Belarus/RSNA images and probes are tested on the remaining 40%. INLP refits a logistic probe and projects onto its null space each round, for up to 200 rounds. LEACE is the closed-form binary-concept eraser. Zero-variance feature columns are excluded from the fit and passed through, because including them lets the erased direction leak back into those columns in the 1024-d XRV features, which left a linear probe at AUC 0.997. In our diagnostic, 108 of the 1024 XRV columns had zero variance (covariance rank 916) and the largest standardized test value was 931 standard deviations, and LEACE on the full space left a linear probe at train AUC 1.000 although all class means were equal to 10^{-11}. Dropping the dead columns gave train 0.509 and test 0.524, and clipping outliers made no difference. The diagnostic is consistent with the erased direction leaking into the dead columns at floating-point level and being re-amplified by the probe's standardization. The planted-marker runs apply LEACE after a 64-d PCA and are not affected.

Ablation. For a unit direction $\hat{d}$ the ablated feature is $z - (z^\top \hat{d})\hat{d}$. The shortcut direction d_s is the weight vector of a logistic probe for s, the classifier direction d_c is the weight vector of the logistic classifier, and the residual direction d_r is d_c minus its projection on d_s, normalized. Each ablation is evaluated with the original head (fixed), with a head retrained on ablated features, and against 200 covariance-matched random directions, from which an empirical p-value is computed. The number of seeds is 3 for TB XRV, 2 for TB DINOv2, 5 for malaria, and 10 for glioma.

Malaria shortcut and views. The staining shortcut is a high-versus-low mean saturation split. The parasite region keeps dilated mask pixels, cropped around the center. The RBC view replaces dilated mask pixels by the median color of the remaining cell pixels. The random crop is a circle of radius 35% of the smaller image side, off-center, chosen from up to 50 candidates for least overlap with the dilated mask (mean overlap 1.2%, 28.95% of crops with any overlap). Grayscale is standardized within the cell, and the silhouette keeps only the cell outline. Handcrafted features are color statistics and size-and-shape statistics. Masks are approximate thresholds (a fixed HSV rule or a per-cell adaptive rule, depending on the encoder run) and were not validated against annotations. A coarser P# grouping with ResNet-50 gave the same patient-grouped AUCs within 0.001 to 0.002 for every view.

Planted marker. A bright corner block with a soft gradient is added to Shenzhen and Montgomery radiographs with $P(\text{marker} \mid \text{TB}) = \rho$, so ρ is the marker prevalence among TB cases, $1 - \rho$ the prevalence among normals, $\rho = 0.5$ is the no-association setting, and $\rho = 1$ gives perfect association. Features are reduced to a 64-d PCA space and the procedure is repeated over 20 splits per ρ. The audit flags reliance when LEACE on the recovered marker direction raises AUC on marker-flipped test images by more than 0.02 and above the 95th percentile of the random-direction null gain. The shortcut tax is concordant AUC minus flipped AUC.

Statistics. Bracketed ranges are 95% intervals. Intervals are percentile bootstrap intervals over test images, or over patients in the patient-level tests, and exclude training variance. Intervals for malaria view AUCs resample cells and ignore patient clustering, so they are too narrow. The RBC-versus-random-crop comparison is a paired bootstrap against an equivalence margin of ±0.02.

Appendix E. Index of Controls

Table 1 maps every control to where it is reported, so a reader can find the full result behind each statement in the main text.

Table 1: Controls and where they are reported.

Control	Location
Retrained head, random-direction null	Table 4
Planted-marker calibration	Section 5.1, Tables 13 and 14
Source-matched TB transfer, metadata-only baseline	Table 2
Linear, MLP, and HGB probes after erasure	Table 3, Figure 3a
Harmonized TB variant	Tables 2 and 3
Eraser transport, head inheritance, CKA	Tables 15, 16 and 17
Glioma refits, imputation variants, label-permutation null	Tables 5 and 6
Glioma grade ablation	Tables 4 and 7
Patient-grouped malaria splits, three encoders	Table 8
RBC and other views, mask overlap	Table 8, Appendix D
Bystander and patient-identifiability tests	Tables 9, 10, 11 and 12

Appendix F. TB: Extended Results

Source-matched transfer. Within-source classification is strong (Shenzhen 0.893 to 0.903 and Montgomery 0.844 to 0.903 across the three unharmonized encoders, Table 2). Transfer between the source-matched datasets is partial. Shenzhen to Montgomery gives 0.628 to 0.736, and Montgomery to Shenzhen gives 0.787 (DINOv2) and 0.821 (ResNet-50) but 0.356 for XRV, below chance, which remains unexplained. The Belarus/RSNA-trained classifier scores 0.395 (DINOv2), 0.465 (ResNet-50) and 0.501 (XRV) on Shenzhen alone, and 0.569, 0.690 and 0.673 on Montgomery alone, so its failure is concentrated on Shenzhen. The cosine between the source direction and the direction of a classifier trained on source-matched data is 0.045, 0.031 and 0.002 for DINOv2, ResNet-50 and XRV, which is about the alignment expected between random directions in 384 to 2048 dimensions (0.02 to 0.05), so the source direction shows no alignment with the direction the source-matched classifier uses.

Metadata and harmonization. Image size and intensity statistics alone separate Belarus from RSNA at AUC 1.000 and predict TB within the source-matched data at 0.748 [0.713, 0.780], so image-level statistics separate TB from normal even there, although intensity statistics may partly encode pathology. Harmonization leaves source decodable (0.999) and raises the Belarus/RSNA-trained classifier on source-matched data from 0.542 to 0.752 for XRV, 0.802 on Shenzhen and 0.705 on Montgomery. Because harmonization changes format while population differences remain, this shows format contributes to the failure and does not show that format is the only cause.

Erasure. LEACE reaches linear chance in one step (0.470 to 0.494 for the four feature sets). INLP needs 100 rounds for DINOv2 (0.496) and 200 for ResNet-50 (0.484), XRV (0.501) and harmonized XRV (0.503) (Figure 3a, Table 3). After 200 rounds MLP probes still reach 0.941 (DINOv2) and at least 0.990 for the others, and HGB probes reach 0.831 (DINOv2) and at least 0.991 for the others. Cross-validated TB classification within the source-matched data is unchanged by erasure (None versus LEACE: DINOv2 0.839 vs. 0.844, ResNet-50 0.850 vs. 0.847, XRV 0.808 vs. 0.825, harmonized 0.812 vs. 0.827), so erasing the linear source direction does not remove information that classification within the source-matched data needs.

Extrapulmonary-involvement ablation. The extrapulmonary direction is fit within TB-positive cases (group A versus group B), so the test addresses reliance on extrapulmonary involvement within cases and not the TB versus normal decision. In TB XRV features, ablating this direction leaves positive-class accuracy near baseline but lowers normal accuracy from 0.998 to 0.835 ($p = 0.01$), and a retrained head restores 0.998 (Table 4). In DINOv2 features, the same ablation changes nothing (overall 0.999 to 0.998, $p = 0.92$). The residual ablation with a fixed head zeroes positive-class accuracy in DINOv2 (0.000 for both groups) and lowers it to 0.108 and 0.236 in XRV, while retrained heads return to 0.976 to 0.989, so the information was

still present in the ablated features and the fixed-head drop reflects the head's dependence on the removed direction.

Table 2: TB source-matched evaluation (AUC with 95% bootstrap interval over test images). Metadata (image size and intensity statistics) alone separate Belarus from RSNA at AUC 1.000 and predict TB within Shenzhen and Montgomery at AUC 0.748 [0.713, 0.780].

Test	DINOv2	ResNet-50	XRV	harmonized XRV
Source probe, Belarus vs. RSNA	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.999 [0.999, 1.000]	0.999 [0.999, 1.000]
Within Shenzhen (CV)	0.893 [0.869, 0.916]	0.897 [0.873, 0.920]	0.903 [0.879, 0.925]	0.930 [0.910, 0.948]
Within Montgomery (CV)	0.844 [0.775, 0.906]	0.870 [0.809, 0.922]	0.903 [0.840, 0.954]	0.898 [0.829, 0.951]
Shenzhen → Montgomery	0.736 [0.652, 0.820]	0.718 [0.624, 0.802]	0.628 [0.528, 0.722]	0.548 [0.446, 0.645]
Montgomery → Shenzhen	0.787 [0.753, 0.820]	0.821 [0.791, 0.851]	0.356 [0.318, 0.398]	0.608 [0.567, 0.650]
Belarus/RSNA → Shenzhen+Montgomery	0.455 [0.417, 0.499]	0.511 [0.471, 0.552]	0.542 [0.503, 0.584]	0.752 [0.715, 0.785]
Belarus/RSNA → Shenzhen	0.395 [0.350, 0.436]	0.465 [0.418, 0.507]	0.501 [0.455, 0.545]	0.802 [0.767, 0.835]
Belarus/RSNA → Montgomery	0.569 [0.466, 0.670]	0.690 [0.598, 0.779]	0.673 [0.573, 0.762]	0.705 [0.609, 0.792]
Cosine(source dir., source-matched classifier dir.)	0.045	0.031	0.002	0.001

Table 3: Held-out source AUC after erasure (linear / MLP / HGB probes) and source-matched TB classification AUC (cross-validated on Shenzhen+Montgomery, "cls."). Erasers are fit on 60% of the Belarus/RSNA images and probes are tested on the remaining 40%. n/a means the quantity was not computed.

	DINOv2				ResNet-50				XRV				harmonized XRV			
Eraser	lin.	MLP	HGB	cls.	lin.	MLP	HGB	cls.	lin.	MLP	HGB	cls.	lin.	MLP	HGB	cls.
None	1.000	1.000	1.000	0.839	1.000	1.000	0.999	0.850	0.999	0.999	1.000	0.808	0.999	0.999	1.000	0.812
LEACE	0.494	0.998	0.985	0.844	0.470	0.997	0.995	0.847	0.484	0.989	1.000	0.825	0.481	0.998	1.000	0.827
INLP 1	0.999	0.999	0.999	n/a	0.999	1.000	0.999	n/a	0.999	0.994	0.990	n/a	0.999	0.999	0.992	n/a
INLP 10	0.967	0.998	0.990	n/a	0.994	0.999	0.999	n/a	0.988	0.991	1.000	n/a	0.988	0.999	1.000	n/a
INLP 50	0.671	0.996	0.977	0.831	0.905	0.999	0.998	0.854	0.882	0.996	1.000	0.825	0.901	0.999	1.000	0.831
INLP 100	0.496	0.989	0.953	n/a	0.788	0.999	0.997	n/a	0.684	0.999	1.000	n/a	0.658	0.999	0.999	n/a
INLP 200	0.472	0.941	0.831	n/a	0.484	0.994	0.991	n/a	0.501	0.990	0.999	n/a	0.503	0.991	0.995	n/a

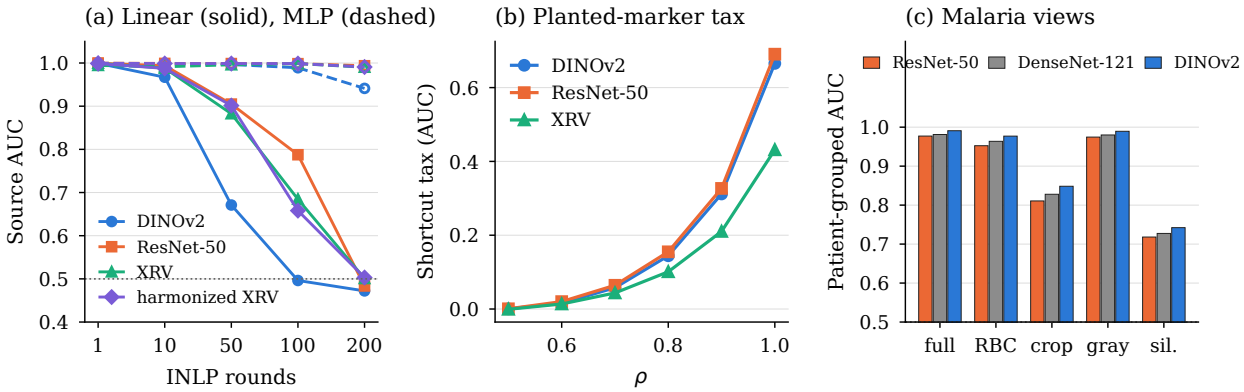


Figure 3: (a) Source AUC after INLP by number of rounds (linear probe solid, MLP probe dashed), for four feature sets. (b) Planted-marker shortcut tax (concordant minus flipped AUC) by ρ. (c) Malaria patient-grouped AUC by view and encoder (chance 0.5).

Table 4: Ablation controls, accuracy (empirical $p = (1 + k)/(N + 1)$ in parentheses, where k of $N = 200$ covariance-matched random directions change accuracy at least as much as the ablation, so the smallest attainable value is 0.005 and is shown as <0.005). Seeds: TB XRV 3, TB DINOv2 2, malaria 5, glioma 10. Malaria shortcut is high vs. low mean saturation. Single-class rows carry little information on their own. TB uses the extrapulmonary-involvement direction fit within TB-positive cases. GBM denotes glioblastoma, LGG lower-grade glioma and wt wildtype. For glioma the grade direction has cosine 1.00 with the classifier, so the residual direction is degenerate and the residual columns repeat the shortcut columns.

Domain	Subgroup	Baseline	Shortcut, fixed head	Shortcut, retrained	Residual, fixed head	Residual, retrained
TB XRV	Group A positives	0.991	0.991 (0.79)	0.989 (1.00)	0.108 (<0.005)	0.989 (1.00)
TB XRV	Group B positives	0.986	0.990 (0.56)	0.988 (1.00)	0.236 (<0.005)	0.986 (0.74)
TB XRV	Normals	0.998	0.835 (0.01)	0.998 (1.00)	0.290 (<0.005)	0.998 (0.74)
TB XRV	All	0.997	0.851 (0.01)	0.997 (0.97)	0.278 (<0.005)	0.997 (1.00)
TB DINOv2	Group A positives	1.000	1.000 (1.00)	1.000 (1.00)	0.000 (<0.005)	0.986 (0.05)
TB DINOv2	Group B positives	0.991	0.991 (1.00)	0.991 (1.00)	0.000 (<0.005)	0.976 (0.10)
TB DINOv2	Normals	0.999	0.999 (0.79)	0.999 (1.00)	1.000 (0.60)	0.997 (0.03)
TB DINOv2	All	0.999	0.998 (0.92)	0.999 (1.00)	0.894 (0.02)	0.995 (0.03)
Malaria full	All	0.928	0.927 (0.95)	0.928 (0.45)	0.506 (0.01)	0.931 (0.03)
Malaria full	Parasitized	0.922	0.922 (0.95)	0.923 (0.65)	0.200 (<0.005)	0.924 (0.03)
Malaria full	Uninfected	0.934	0.934 (0.93)	0.935 (0.58)	0.800 (0.38)	0.939 (0.03)
Malaria RBC	All	0.890	0.886 (0.81)	0.890 (0.74)	0.504 (<0.005)	0.893 (0.03)
Malaria RBC	Parasitized	0.879	0.880 (0.93)	0.879 (0.87)	0.328 (<0.005)	0.879 (0.03)
Malaria RBC	Uninfected	0.902	0.893 (0.91)	0.902 (0.84)	0.703 (0.28)	0.908 (0.06)
Glioma	Concordant	1.000	0.490 (0.12)	0.815 (0.11)	0.490 (0.12)	0.815 (0.11)
Glioma	Discordant IDH-wt LGG	0.000	0.000 (1.00)	0.654 (0.06)	0.000 (1.00)	0.654 (0.06)
Glioma	Discordant IDH-mut GBM	0.000	1.000 (0.06)	0.622 (0.03)	1.000 (0.06)	0.622 (0.03)

Appendix G. Glioma: Extended Results

Grade ablation. The grade direction has cosine 1.00 with the classifier direction (Table 7), so the residual direction is degenerate (a zero vector, so the residual columns of Table 4 repeat the shortcut columns) and ablating the grade direction removes the classifier's only informative input. The ablated classifier predicts IDH-mutant for all 398 test patients. Accuracy on the concordant set (0.490) is below the 0.510 obtained by predicting all-wildtype, and the apparent gain on IDH-mutant glioblastoma (1.000) is the all-mutant prior. The fixed-head and retrained-head accuracies for the three subgroups are in Table 4, where a retrained head reaches 0.815, 0.654 and 0.622 on the concordant, IDH-wildtype lower-grade, and IDH-mutant glioblastoma groups.

Refits without grade. Table 5 lists every variant. The refit with zero imputation reaches discordant AUC 0.738, median imputation 0.812, and median imputation with missingness flags 0.563 (mean over 20 splits; the bootstrap AUC is 0.502, with interval [0.394, 0.596]), so the estimate depends on analyst choices. Dropping 1p/19q lowers it to 0.624 and dropping 1p/19q and TERT gives 0.638, identical to the MGMT-only model since MGMT is then the only input. TERT alone is below chance on the discordant set (0.336), and 1p/19q alone is 0.527. The label-permutation null means are 0.422 to 0.528, with SDs of 0.13 to 0.29 across splits (0.03 for 1p/19q alone, a single sparse feature), because each permuted-label model is a random function scored on 141 structured discordant patients and can land far from 0.5 in either direction. With 20 splits the means therefore lie within two standard errors of 0.5 (largest $|z| = 1.95$, for median imputation with missingness flags: mean 0.422, SD 0.179, so SE 0.040), and the variants reuse the same splits and permutation seeds, so the rows are not independent. Missingness indicators alone predict IDH at 0.666 [0.631, 0.699] (Table 6), so part of any refit signal can come from which assays were run and not from their values.

Stratified by grade. With grade held constant within a stratum, the refit reaches 0.891 [0.855, 0.925] in lower-grade glioma and 0.606 [0.509, 0.704] in glioblastoma, and without 1p/19q and TERT 0.750 and 0.604. The within-glioblastoma signal is therefore weak. Because 1p/19q co-deletion implies IDH mutation by definition (Louis et al., 2021), the within-lower-grade value is partly definitional.

Table 5: Glioma refits (20 splits). Discordant AUC (mean over splits) with a 95% bootstrap interval, and mean of the label-permutation null. The interval is centered on a separate bootstrap point estimate, which differs from the mean only for the median+missing flags row (0.502 vs. 0.563).

Variant	Conc. AUC	Disc. AUC	Perm. null
full (with grade)	1.000	0.000 [0.000, 0.000]	0.476
no grade, zero-impute	0.877	0.738 [0.649, 0.823]	0.452
no grade, median-impute	0.806	0.812 [0.736, 0.888]	0.428
no grade, median+missing flags	0.948	0.563 [0.394, 0.596]	0.422
no grade, no 1p/19q	0.834	0.624 [0.544, 0.716]	0.464
no grade, no 1p/19q, no TERT	0.809	0.638 [0.548, 0.723]	0.528
1p/19q only	0.698	0.527 [0.500, 0.566]	0.500
TERT only	0.581	0.336 [0.284, 0.387]	0.484
MGMT only	0.809	0.638 [0.548, 0.723]	0.528

Table 6: Glioma robustness. Cross-validated AUC with 95% bootstrap interval for grade-stratified refits (GBM denotes glioblastoma, LGG lower-grade glioma) and for missingness indicators alone.

Setting	AUC [95% CI]
no grade, zero-impute | LGG only	0.891 [0.855, 0.925]
no grade, zero-impute | GBM only	0.606 [0.509, 0.704]
no grade, no 1p/19q, no TERT | LGG only	0.750 [0.686, 0.808]
no grade, no 1p/19q, no TERT | GBM only	0.604 [0.510, 0.703]
Missingness indicators only	0.666 [0.631, 0.699]

Table 7: Glioma grade ablation: fraction predicted IDH-mutant (accuracy in parentheses) per test subgroup. The ablated classifier predicts IDH-mutant for everyone. Majority-class accuracy on the concordant set is 0.510. Cosine between the grade direction and the classifier direction: 1.00. GBM denotes glioblastoma and LGG lower-grade glioma.

Subgroup	n	Baseline	Grade ablated
Concordant	257	0.490 (1.000)	1.000 (0.490)
Disc. IDH-wt LGG	104	1.000 (0.000)	1.000 (0.000)
Disc. IDH-mut GBM	37	0.000 (0.000)	1.000 (1.000)

Appendix H. Malaria: Extended Results

Splitting. Patient-grouped AUC is 0.002 to 0.005 below cell-level AUC for the full image and 0.002 to 0.033 below it across all views (Table 8). In total, 24.9% of patient groups, holding 12.4% of cells, contain a single class. These groups offer no within-patient contrast, so for them patient identity and label coincide.

Views. Across the three encoders the full image reaches 0.977 to 0.991, grayscale 0.975 to 0.989, the RBC view 0.952 to 0.977, the parasite region alone 0.899 to 0.928, the random crop 0.811 to 0.848, and the silhouette 0.718 to 0.742 (Figure 3c). Size-and-shape features alone reach 0.575 to 0.577, color-only features 0.967 to 0.969, and all handcrafted features except the parasite fraction 0.971 to 0.972. The RBC view minus the random crop is 0.142 for ResNet-50, 0.136 for DenseNet-121, and 0.129 for DINOv2, each with an interval that excludes the ± 0.02 equivalence margin. With a mean mask overlap of 1.2% the random crop contains little parasite, so the gap is not caused by the crop containing parasite pixels. It could reflect crop size or position, so it is not an estimate of how much information lies outside the parasite.

Staining ablation. A saturation-defined staining direction ablated with a fixed head leaves ResNet-50 accuracy at 0.927 against 0.928 at baseline ($p = 0.95$), and in the RBC view at 0.886 against 0.890 ($p = 0.81$) (Table 4). The residual ablation with a fixed head reduces accuracy to 0.506 and 0.504 and a retrained head returns it to 0.931 and 0.893. A direction defined by saturation is one proxy for staining and excludes other stain properties, so the null result does not establish that stain is irrelevant.

Patient state. Uninfected cells from patients with at least one infected cell (bystander cells; 10,364 cells from 151 patients) have higher infected scores than uninfected cells from the 50 patients with none (uninfected-patient cells; 3,415 cells), with AUC 0.612 to 0.675 for full-image features and 0.505 to 0.512 for the parasite region (Table 9). In ResNet-50 the false-positive rate is 0.068 for bystander cells and 0.040 for

uninfected-patient cells (Table 11). Patient infection status can be predicted from uninfected cells alone at patient-level AUC 0.946 to 0.954 for full-image features, and a probe identifies the source patient among 201 at top-1 accuracy 0.499 to 0.582 for full images and 0.491 to 0.582 for the RBC view, against a chance of 0.005 (Table 10). The silhouette alone identifies the patient at 0.068 to 0.098, more than ten times chance. In ResNet-50 the bystander excess false-positive rate (bystander minus uninfected-patient) is 0.135 for the random crop and 0.177 for the silhouette against 0.028 for the full image (Table 12), so the patient-state cue is not confined to parasite pixels. These results are compatible with slide-level or patient-level appearance, which cell-level splitting would leak, but slide metadata are unavailable, so the cue is not identified.

Table 8: Malaria, patient-grouped (5-fold, 201 groups) and cell-level cross-validated AUC with 95% intervals. RBC view minus random crop (patient-grouped, paired bootstrap): ResNet-50 0.142 [0.137, 0.146], DenseNet-121 0.136 [0.131, 0.140], DINOv2 0.129 [0.124, 0.133]. The ±0.02 equivalence test failed for all three. Intervals resample cells and ignore patient clustering, so they are too narrow.

	Patient-grouped			Cell-level		
View	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full image	0.977 [0.975, 0.978]	0.981 [0.980, 0.982]	0.991 [0.990, 0.992]	0.982 [0.980, 0.983]	0.986 [0.985, 0.987]	0.993 [0.992, 0.993]
Parasite region	0.899 [0.895, 0.904]	0.917 [0.914, 0.921]	0.928 [0.925, 0.931]	0.907 [0.903, 0.911]	0.926 [0.923, 0.929]	0.936 [0.934, 0.940]
RBC view	0.952 [0.950, 0.954]	0.964 [0.962, 0.966]	0.977 [0.975, 0.978]	0.964 [0.963, 0.966]	0.974 [0.972, 0.975]	0.982 [0.980, 0.983]
Random crop	0.811 [0.805, 0.815]	0.828 [0.823, 0.833]	0.848 [0.844, 0.852]	0.844 [0.839, 0.849]	0.860 [0.855, 0.864]	0.874 [0.870, 0.877]
Grayscale	0.975 [0.973, 0.976]	0.980 [0.979, 0.981]	0.989 [0.988, 0.990]	0.978 [0.977, 0.980]	0.984 [0.983, 0.985]	0.991 [0.990, 0.992]
Silhouette	0.718 [0.713, 0.723]	0.727 [0.721, 0.732]	0.742 [0.737, 0.747]	0.726 [0.721, 0.731]	0.737 [0.731, 0.743]	0.751 [0.745, 0.756]
Color-only features	0.968 [0.966, 0.970]	0.969 [0.967, 0.970]	0.967 [0.965, 0.969]	0.982 [0.981, 0.984]	0.983 [0.982, 0.984]	0.983 [0.982, 0.984]
Size and shape only	0.575 [0.568, 0.582]	0.576 [0.569, 0.583]	0.577 [0.570, 0.583]	0.586 [0.579, 0.593]	0.585 [0.578, 0.591]	0.588 [0.581, 0.595]
All handcrafted, no parasite fraction	0.971 [0.969, 0.973]	0.972 [0.970, 0.973]	0.971 [0.969, 0.973]	0.985 [0.984, 0.986]	0.985 [0.984, 0.986]	0.985 [0.983, 0.986]

Table 9: Malaria patient-state test. (i) AUC separating uninfected cells of patients with at least one infected cell (10,364 cells) from uninfected cells of patients with none (3,415 cells), using the infected-vs.-uninfected score of a patient-held-out classifier. (ii) Patient-level AUC for predicting whether a patient has any infected cell from uninfected cells alone (scores averaged per patient, 95% cluster bootstrap over patients).

	(i) Bystander vs. uninfected-patient AUC			(ii) Patient-level status AUC		
View	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full	0.644	0.675	0.612	0.954 [0.909, 0.987]	0.949 [0.904, 0.982]	0.946 [0.903, 0.981]
Parasite	0.505	0.505	0.512	0.796 [0.721, 0.857]	0.875 [0.821, 0.922]	0.886 [0.825, 0.938]
RBC	0.669	0.678	0.651	0.953 [0.910, 0.988]	0.948 [0.902, 0.980]	0.949 [0.904, 0.981]
Random crop	0.665	0.690	0.699	0.911 [0.861, 0.954]	0.930 [0.881, 0.969]	0.941 [0.894, 0.977]
Gray	0.568	0.592	0.608	0.934 [0.883, 0.967]	0.948 [0.909, 0.978]	0.909 [0.846, 0.952]
Silhouette	0.645	0.662	0.660	0.886 [0.830, 0.935]	0.874 [0.812, 0.928]	0.854 [0.788, 0.909]

Table 10: Malaria patient-identifiability probe: top-1 accuracy of identifying the source patient among 201 from a held-out cell (chance 0.005).

View	ResNet-50	DenseNet-121	DINOv2
Full image	0.499	0.582	0.567
RBC view	0.491	0.582	0.567
Random crop	0.330	0.367	0.428
Grayscale	0.336	0.402	0.395
Silhouette	0.068	0.074	0.098

Table 11: Malaria bystander test, full-image features. False-positive rate of the infected-vs.-uninfected classifier on uninfected cells from patients with (bystander) and without (uninfected-patient) an infected cell, and cell-level AUC of predicting patient status from uninfected cells. 201 patients, 151 with an infected cell.

Quantity	ResNet-50	DenseNet-121	DINOv2
FPR, bystander	0.068	0.056	0.040
FPR, uninfected-patient	0.040	0.034	0.028
Cell-level status AUC	0.885	0.900	0.903

Table 12: Malaria bystander test for every view: false-positive rate of the infected-vs.-uninfected classifier on uninfected cells from patients with an infected cell (bystander) and from patients with none (uninfected-patient), and cell-level AUC of predicting patient status from uninfected cells.

	FPR bystander			FPR uninfected-patient			Cell-level status AUC		
View	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2	ResNet-50	DenseNet-121	DINOv2
Full	0.068	0.056	0.040	0.040	0.034	0.028	0.885	0.900	0.903
Parasite	0.075	0.067	0.059	0.073	0.060	0.050	0.537	0.556	0.590
RBC	0.108	0.083	0.073	0.052	0.044	0.039	0.886	0.900	0.905
Random crop	0.246	0.227	0.192	0.111	0.097	0.088	0.837	0.874	0.895
Gray	0.079	0.060	0.044	0.048	0.043	0.029	0.833	0.869	0.842
Silhouette	0.364	0.364	0.352	0.187	0.164	0.158	0.714	0.741	0.739

Appendix I. Planted Marker: Extended Results

The flag rate is 0.00 at $\rho = 0.5$ for all three encoders (Table 13). For DINOv2 it is 0.40, 0.85, 1.00, 1.00 and 0.90 for $\rho = 0.6$ to 1.0, for ResNet-50 0.25, 0.85, 1.00, 1.00 and 1.00, and for XRV 0.00, 0.30, 0.75, 0.70 and 0.00. The shortcut tax rises with ρ for all encoders (Figure 3b) and is smaller for XRV (0.431 at $\rho = 1.0$ against 0.666 and 0.691). The recovered direction has cosine 0.64 to 0.69 with the planted direction for DINOv2 and 0.66 to 0.72 for ResNet-50, but 0.25 to 0.27 for XRV, so the recovered XRV direction is only weakly aligned with the marker.

At $\rho = 1.0$ the planted marker is perfectly associated with the label by construction. After erasure, a retrained head scored AUC 0.517 for DINOv2, 0.507 for ResNet-50 and 0.540 for XRV on marker-flipped test images. For XRV the LEACE gain at $\rho = 1.0$ is negative (-0.182, Table 14), which is why the flag rate falls to 0.00 there, and for DINOv2 it is 0.142 with flag rate 0.90. A flag at $\rho = 1.0$ is therefore not guaranteed, and a clean audit at very high marker prevalence should not be read as absence of the marker. After LEACE the marker presence AUC from the residual is near 0.5 (0.510 for ResNet-50 at $\rho = 0.7$, Table 14), so the eraser removes the linearly decodable marker.

Table 13: Planted-marker calibration (20 splits per ρ): flag rate, shortcut tax (concordant minus flipped AUC), cosine between recovered and planted direction (in a 64-d PCA space), and AUC on flipped images after erasing the marker and retraining the head.

	DINOv2				ResNet-50				XRV			
ρ	flag	tax	cos	retr.	flag	tax	cos	retr.	flag	tax	cos	retr.
0.5	0.00	0.001	0.68	0.867	0.00	0.001	0.72	0.881	0.00	-0.001	0.26	0.895
0.6	0.40	0.016	0.69	0.869	0.25	0.020	0.72	0.882	0.00	0.014	0.27	0.887
0.7	0.85	0.058	0.69	0.878	0.85	0.064	0.71	0.890	0.30	0.044	0.26	0.887
0.8	1.00	0.144	0.69	0.889	1.00	0.155	0.70	0.901	0.75	0.101	0.26	0.884
0.9	1.00	0.312	0.67	0.895	1.00	0.327	0.69	0.913	0.70	0.210	0.25	0.864
1.0	0.90	0.666	0.64	0.517	1.00	0.691	0.66	0.507	0.00	0.431	0.25	0.540

Table 14: Planted marker, additional quantities (mean over 20 splits): AUC on concordant and flipped test images, gain in flipped AUC from LEACE on the marker, 95th percentile of the random-direction null gain, and AUC of marker presence from the LEACE residual (0.5 means the marker is no longer linearly decodable).

	DINOv2					ResNet-50					XRV				
ρ	conc.	flip	gain	p95	resid.	conc.	flip	gain	p95	resid.	conc.	flip	gain	p95	resid.
0.5	0.868	0.867	-0.000	0.002	0.486	0.884	0.883	-0.002	0.002	0.496	0.896	0.896	-0.002	0.002	0.492
0.6	0.872	0.857	0.013	0.002	0.494	0.891	0.871	0.009	0.002	0.515	0.901	0.887	-0.003	0.004	0.497
0.7	0.885	0.827	0.049	0.003	0.496	0.904	0.840	0.045	0.003	0.510	0.911	0.867	0.011	0.010	0.508
0.8	0.901	0.757	0.123	0.006	0.504	0.923	0.768	0.122	0.008	0.491	0.925	0.824	0.043	0.020	0.498
0.9	0.928	0.616	0.250	0.014	0.492	0.949	0.622	0.258	0.020	0.508	0.943	0.733	0.078	0.045	0.506
1.0	0.967	0.301	0.142	0.035	0.496	0.985	0.295	0.203	0.049	0.514	0.969	0.538	-0.182	0.084	0.515

Appendix J. Eraser Transport, Head Inheritance, and CKA

Transport. The transport experiments are a separate run from the erasure experiments above, so their own-eraser and native-head values differ from Tables 2 and 3 by up to 0.06 (for example, own-LEACE linear

source AUC in ResNet-50 0.529 against 0.470). We fit LEACE in encoder A, map its direction to encoder B through a ridge map, and apply it there, with a random rank-1 eraser as control (Table 15). Transported erasers reduce the linear source probe less than an eraser fit in B itself (DINOv2 to ResNet-50 gives 0.931 against 0.529, and the reverse gives 0.997 against 0.511), the random rank-1 control leaves it at 0.998 to 1.000, and MLP probes decode source at 0.991 or above in every transported condition. The held-out R^2 of the ridge maps is 0.14 to 0.36 (Table 16), so the maps are loose and the result tests practical transfer with approximate maps.

Head inheritance. A Belarus/RSNA-trained head from A applied through the map in B scores AUC 0.467 (ResNet-50 from DINOv2), 0.463 (DINOv2 from ResNet-50), 0.499 (XRV from DINOv2) and 0.541 (DINOv2 from XRV) on source-matched data, within the range of the native heads (0.464 to 0.570 here, 0.455 to 0.542 in Table 2), and its scores correlate 0.757 to 0.897 with the native B head. The confounded classifier is therefore inherited together with the representation, and source-matched TB classification in B is not changed by the own or the transported eraser (Table 16).

CKA. Linear CKA (Kornblith et al., 2019) between XRV and DINOv2 is 0.202 and between ResNet-50 and DINOv2 0.408. Erasing the source direction removes 3.2% to 4.5% of the variance and takes the linear source AUC from 1.000 to 0.471 to 0.493, but CKA between an encoder and its erased version stays at 0.890 (ResNet-50), 0.930 (DINOv2) and 0.978 (XRV), and cross-encoder CKA falls by 0.055 and 0.045, against 0.00017 for isotropic random rank-1 erasures (Table 17). The fall is 263 to 315 times the random value, but the random directions were not matched to the 3.2% to 4.5% of variance that the source direction carries, so the ratio is descriptive. In absolute terms CKA changes little while linear source decodability falls to chance, so a similarity score near 0.9 does not clear a representation of a decodable confound (Harvey et al., 2024; Davari et al., 2023). Representational similarity should be paired with probes.

Table 15: Eraser transport. A LEACE eraser fit in encoder A is mapped to encoder B through a ridge map. Held-out linear source AUC in B (MLP in parentheses). This is a separate run, so own-LEACE values differ from Table 3 by up to 0.06.

A → B	B own LEACE	Transported	Random rank-1
DINOv2 → ResNet-50	0.529 (0.986)	0.931 (0.997)	1.000 (0.999)
ResNet-50 → DINOv2	0.511 (0.991)	0.997 (0.994)	1.000 (0.999)
DINOv2 → XRV	0.524 (0.987)	0.987 (0.991)	0.998 (0.999)
XRV → DINOv2	0.511 (0.991)	0.991 (0.993)	1.000 (0.999)

Table 16: Eraser transport and head inheritance on source-matched data (Shenzhen+Montgomery). A ridge map takes encoder A features to encoder B. Held-out R^2 of the map in each direction, AUC of a Belarus/RSNA-trained head from A applied through the map in B (stitched), AUC of the native heads, correlation of stitched and native B scores, and source-matched TB classification AUC in B with no eraser, B's own LEACE and the transported LEACE. This is a separate run, so native-head AUCs differ from Table 2 by up to 0.05.

A → B	R^2 A→B	R^2 B→A	Stitched head	Native head B	Native head A	Corr.	Cls. none	Cls. own	Cls. transp.
DINOv2 → ResNet-50	0.357	0.323	0.467 [0.428, 0.509]	0.464 [0.424, 0.504]	0.471 [0.433, 0.514]	0.78	0.850	0.848	0.844
ResNet-50 → DINOv2	0.323	0.357	0.463 [0.426, 0.503]	0.471 [0.433, 0.514]	0.464 [0.424, 0.504]	0.90	0.839	0.843	0.836
DINOv2 → XRV	0.273	0.140	0.499 [0.461, 0.540]	0.570 [0.531, 0.611]	0.471 [0.433, 0.514]	0.76	0.808	0.815	0.813
XRV → DINOv2	0.140	0.273	0.541 [0.503, 0.582]	0.471 [0.433, 0.514]	0.570 [0.531, 0.611]	0.84	0.839	0.843	0.847

Table 17: Linear centered kernel alignment (CKA) before and after erasing the TB source direction with LEACE (A is the first encoder, B the second). Random rank-1 is the mean over random rank-1 erasures. This is a separate run, so source AUCs differ from Tables 3 and 15 in the third decimal place.

Pair	Source AUC A, A erased	Source AUC B, B erased	CKA(A,B)	CKA(A, A erased)	CKA(B, B erased)	Variance removed A / B	Decrease in CKA(A,B)	Random rank-1 decrease
ResNet-50 / DINOv2	1.000, 0.471	1.000, 0.493	0.408	0.890	0.930	0.045 / 0.043	0.055	0.00017
XRV / DINOv2	1.000, 0.481	1.000, 0.493	0.202	0.978	0.930	0.032 / 0.043	0.045	0.00017

Appendix K. Deviations and Analysis Status

- **Pre-registration.** Decision thresholds such as the 0.02 flag margin were set in our analysis plan and were not registered externally.

- **Follow-up selection.** Follow-up tests were chosen after earlier results. No multiplicity correction was applied.
- **Glioma cohort.** The cohort contains the 995 patients with IDH status, which is 1,122 rows minus 127 rows without IDH status.
- **Glioma grade ablation.** Ablating the grade direction yields a constant prediction (Section 5.3, Table 7) and carries no subgroup information.
- **Malaria staining.** The saturation-defined direction leaves accuracy unchanged (0.927 vs. 0.928), so staining is neither supported nor excluded.
- **Malaria localization.** The RBC-only versus random-crop ordering depends on the mask definition.

Appendix L. Limitations

(1) TB positives and RSNA normals differ in both format and source population, and the source-matched Shenzhen and Montgomery data differ in population again, so format effects are not isolated. Metadata alone predict TB within the source-matched data (AUC 0.748). (2) The XRV encoder is the least sensitive in calibration. (3) The planted marker is a single synthetic corner pattern in TB radiographs, and calibration was not run for glioma or malaria. (4) The glioma domain is tabular, has no age or sex, and the discordant set has 141 patients. (5) Malaria masks are unvalidated, patient groups are parsed from file names (201 groups against the 200 patients the dataset describes), and 24.9% of groups are single-class. (6) Intervals reflect test-sample resampling, with few seeds for ablation controls (2 to 10). (7) Grayscale conversion and the cell silhouette view test complementary aspects of the image, but do not fully isolate staining and background cues. (8) Harmonization was applied to XRV features only, and fine-tuned encoders were not studied. (9) Ablation and erasure act on frozen representations and do not identify causes in the clinical data-generating process. (10) The planted-marker calibration evaluated one flag criterion, so its results may not generalize to other criteria. (11) The TB extent verdict rests on two encoders, XRV and DINOv2, and the XRV calibration shows lower power than DINOv2 and ResNet-50. (12) The 20 splits per marker prevalence resample the same images and are not independent, so the one-sided bound on the false-flag rate (0.14) is optimistic.

Appendix M. Reproducibility

Code, result files and scripts that generate every table and figure are available at https://github.com/rezsam09/Confound-Auditing-in-Medical-Classifiers.